DECISION BRIEF

ARTÉA Phase 3

Solution Overview, Cost–Benefit Analysis, Rationale, Workflow, and Delivery Timeline

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Date: 14 August 2026

Decision: Approve a web-first, reusable-system implementation; keep native iPad/mobile release as a stretch goal

RECOMMENDATION Enhance the existing Godot project rather than rebuild it. Deliver a browser-based minimum viable release first, using shared systems across all three worlds. Defer App Store packaging and multiplayer unless the essential web release is stable by the end of Iteration Two.

Solution Overview

ARTÉA Phase 3 will transform the existing town, Art Castle, Milo companion and three minigames into one guided creative-wellbeing journey. Players enter through a simple web experience, move through Mind Spark, Heart Melody and Creator's Light, collect one treasure from each world, and return to the Castle for the Crown of Life finale. The solution retains the project's established Godot foundation and adds the minimum shared systems needed for progression, reflective guidance, accessibility, reliable saving and browser deployment.

Solution Outline

- Connected journey: the Castle and World Gates provide the main story path; the Crown Train is optional travel to completed worlds only.
- Wonderful Counsellor Milo: a structured, strengths-based response flow—Validate, Reflect, Encourage, Guide and Connect—with bounded prompts and child-safety safeguards.
- Three focused worlds: Mind Spark develops thought awareness; Heart Melody supports emotional expression through music; Creator's Light supports creative expression and reflection.
- Shared technical foundation: one save/progression model, identity-evidence tracker, quest interface, player equipment system, Milo and Guardian scenes, accessibility settings, enemy base and boss-state framework.
- Web-first access: optimise for supported desktop and tablet browsers; treat native iPad/mobile packaging as conditional stretch scope.
- Quality and handover: functional, usability, accessibility and performance testing, followed by source code, setup, deployment and known-issues documentation.

Scope Boundary

Committed Phase 3 scope	Conditional / future scope
Connected three-world journey and Castle finale	Native App Store release after web acceptance
Milo guidance with a remote API and safe fallback responses	Open-ended therapeutic or diagnostic AI behaviour
Shared save, progression, rewards and identity evidence	Multiplayer/community systems
Web deployment, accessibility and performance baseline	Large-scale content expansion or additional worlds
Reuse/modification of existing assets and procedural kits	High-cost bespoke models for every landmark or enemy

Cost Benefit Analysis

FINANCIAL LENS Use the sponsor-cash view for the charter: student labour is an in-kind contribution and therefore a benefit/avoided commercial cost, not a cash expense to the client. A separate full-resource view is shown to avoid implying that labour has no economic value.

Assumptions

- Five team members × 15 hours per week × 14 weeks = 1,050 hours of total student effort.
- Indicative labour value uses the charter's supervisor-approved range of AUD $33–$47 per hour.
- The baseline release is web-first. Native iPad/App Store distribution is excluded from the baseline and requires separate sponsor approval.
- Existing ARTÉA code, world foundations and assets are reused; benefits from reuse are not monetised separately because doing so would double-count the replacement-cost benefit.
- AI estimates assume short, bounded Milo interactions and usage controls. Actual cost depends on model, input/output mix, session length, exchange rate and tax.

Estimated Sponsor Cash Costs (AUD)

Cost category	Basis	Low	High
Godot licence	MIT-licensed engine; no engine licence fee	$0	$0
Milo AI pilot allowance	Bounded text interactions, monitoring and usage cap	$20	$100
Web hosting/backend	Student/free tier where suitable; allowance for paid hosting/storage	$0	$360
Domain / deployment incidentals	Only if a new domain or paid deployment service is required	$20	$40
Testing incidentals	Browser/device access not available through university/team resources	$0	$200
Cash contingency	Reserve for usage variance and deployment issues	$60	$100
Total sponsor cash outlay	Baseline web release	$100	$800

In-Kind Effort and Commercial Replacement Value

Measure	Calculation	Indicative value
Total team effort	5 × 15 hours × 14 weeks	1,050 hours
Low replacement value	1,050 × $33/hour	$34,650
High replacement value	1,050 × $47/hour	$49,350

Measure	Calculation	Indicative value
Client payment for student labour	In-kind academic project contribution	$0 cash

Benefits

Benefit	Value to the sponsor	Treatment in analysis
Commercial development value	Equivalent development, integration, testing, documentation and project-management effort	$34,650–$49,350; monetised
Reuse of existing ARTÉA	Avoids rebuilding the town, Castle, three worlds, player, Milo, art tool and established gameplay foundations	Enables the estimate; not separately monetised
Lower future maintenance	Shared systems reduce duplicated code, asset variants and inconsistent behaviour	Qualitative
Wider access	A browser release can support schools, wellbeing programs and individual sessions without a local installation	Qualitative until usage data exists
Improved engagement	A connected story, visible rewards, Milo guidance and a finale make the experience easier to understand and complete	Qualitative until user testing exists
Service opportunities	Appropriate links to existing courses, workshops and services may generate future engagement	Excluded until conversion data exists

Indicative Net Benefit

Scenario	Benefits	Cash costs	Net benefit	Benefit–cost ratio
Conservative	$34,650	$800	$33,850	43.3 : 1
Midpoint	$42,000	$450	$41,550	93.3 : 1
Upper	$49,350	$100	$49,250	493.5 : 1

Interpretation: the ratio is high because it compares donated/in-kind student development value with the sponsor's incremental cash outlay. It is not a forecast of revenue or social return. In a full economic-resource view, the 1,050 hours are a real resource cost and should not also be counted as a benefit; because wellbeing and service-engagement outcomes are not yet monetised, a full economic benefit–cost ratio should remain TBC until pilot evidence is collected.

Rationale for Chosen Solution

The chosen solution is an incremental, reusable-system enhancement of the existing Godot project with a web-first release. It provides the best balance of client value, delivery certainty and technical sustainability within the 14-week semester. It also preserves the project's therapeutic and faith-informed intent: identity before achievement, discovery before fixing, and celebration before correction.

Option	Cost / schedule	Benefits	Assessment
A. Maintain current version	Lowest immediate cost	No delivery disruption	Rejected: navigation, performance, disconnected activities and limited access remain unresolved.
B. Rebuild as a new platform	Highest cost and schedule risk	Maximum design freedom	Rejected: discards valuable assets/code and is not credible within 14 weeks.
C. Incremental web-first enhancement	Low sponsor cash; manageable technical risk	Reuses current work, connects all worlds, supports testing and enables browser access	Chosen: strongest cost–benefit fit and best chance of an accepted, maintainable release.
D. Native iPad/mobile first	Additional packaging, device and store work	Strong tablet distribution	Stretch only: pursue after web acceptance; Apple distribution adds an annual developer membership and review overhead.

Why the Chosen Solution Is Feasible

1. It reuses the current Godot player, sword hitbox, Milo scene, world foundations, art tool and Crown Line logic instead of replacing them.
2. It builds shared foundations once—save/progression, identity evidence, quest interaction, accessibility, enemy behaviour and boss states—then applies world-specific visuals and puzzle verbs.
3. It narrows the campaign to a guided critical path: Mind Spark provides the largest foundation; Heart Melody and Creator's Light each target 14–16 minutes with two quests and one boss encounter.
4. It protects the essential release by deferring native store packaging, multiplayer, new worlds and open-ended AI behaviour.
5. It makes performance, accessibility and child-safety acceptance criteria part of every iteration rather than end-stage extras.

Design and Safety Rationale

- Thoughts, emotions and creativity are never treated as enemies; combat targets only the fictional blight shell.

- Identity evidence is optional, non-diagnostic and unaffected by combat skill, speed, puzzle accuracy or accessibility settings.
- Milo provides structured reflection and navigation, not diagnosis, crisis counselling or unrestricted therapeutic advice.
- Important cues use text, symbols, shape and sound—not colour alone—and reduced-motion, larger-text, subtitle and help options are included in acceptance testing.

Workflow Diagram

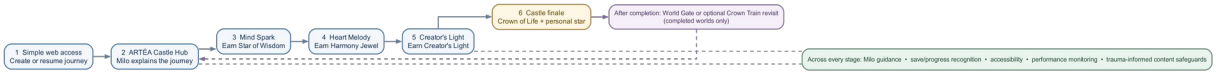


Figure 1. Player and system workflow for the web-first ARTÉA Phase 3 journey.

Milestones and Timelines

The schedule uses five delivery gates. Each gate produces a testable build or approved document, and scope is reduced before quality is reduced. The final acceptance target is 19 October 2026; the charter's earlier 23 October handover date should be corrected because it falls after the stated Week 14 completion date.

Phase / milestone	Dates	Primary output	Exit criteria
1. Charter and technical baseline	3–14 Aug	Approved charter, scoped backlog, repository/build audit	Essential scope, owners, supported browsers and cost assumptions approved
2. Prototype and shared foundations	17–28 Aug	One-hour prototype workshop; web export baseline; save/progression and Milo API spike	Prototype accepted; major technical risks recorded; fallback responses work
3. Iteration One	31 Aug–18 Sep	Shared systems plus playable Mind Spark and Heart Melody critical paths	Iteration One demo accepted; rewards persist; no progression-blocking defects
4. Iteration Two	21 Sep–2 Oct	Creator's Light, Castle finale and integrated three-world journey	End-to-end journey complete in a test build; stretch scope frozen
5. Stabilisation, acceptance and handover	5–19 Oct	Accessibility/performance testing, bug fixes, deployment, video, presentation and handover pack	Supported-browser acceptance passed; source, setup, deployment and known-issues documents delivered

Release Schedule

Release	Target date	Included scope	Decision rule
R0 — Baseline	28 Aug	Current build audit, browser export, Milo/API and performance spikes	If web feasibility is poor, reduce visual/effect load before adding features
R1 — Iteration One	18 Sep	Shared foundations, Milo guidance, Mind Spark and Heart Melody vertical slices	Only defects and essential integration enter the next iteration automatically
R2 — Feature Complete	2 Oct	Creator's Light, rewards, Castle finale and end-to-end save/progression	Freeze multiplayer, App Store and new-content requests unless all essential criteria pass
RC — Release Candidate	16 Oct	Optimised web build, accessibility fixes, test evidence and draft handover	No critical or high-severity defects; sponsor acceptance session completed
Final — Handover	19 Oct	Deployed web build, source, documentation, presentation, abstract/video and known issues	Client/supervisor sign-off or documented residual limitations

Delivery Priorities

Priority	Scope
Must	Connected three-world critical path; Milo guidance and safe fallback; rewards/finale; save/progression; web deployment; accessibility/performance baseline; testing and handover
Should	Optional Crown Train revisits; richer NPC guidance; personal Gallery refinements; service recommendations with sponsor-approved wording
Could	Native iPad packaging if the web release is accepted early and required devices/accounts are available
Won't in Phase 3	Multiplayer/community features, additional worlds, unrestricted generative art recognition, diagnostic AI and large bespoke asset expansion

Measurement at Handover

- Completion: percentage of test users able to finish the three-world critical path without facilitator intervention.
- Reliability: critical/high defects open, crash rate during tests and successful save/resume rate.
- Performance: load time and frame-rate observations on the agreed browser/device matrix.
- Accessibility/usability: navigation errors, help requests and issues found against the agreed accessibility checklist.
- AI cost and quality: average tokens/cost per Milo session, fallback rate and sponsor-reviewed response quality.
- Benefit validation: unique users, repeat sessions, world/finale completion and service referral clicks after pilot launch.

Sources and Calculation Notes

- **ARTÉA World One: Mind Spark — Concise Developer Guide and Implementation Blueprints.** Provided project document, 14 Aug 2026
- **ARTÉA World Two: Heart Melody — Concise Developer Guide and Implementation Blueprints.** Provided project document, 14 Aug 2026
- **ARTÉA World Three: Creator's Light — Concise Developer Guide and Implementation Blueprints.** Provided project document, 14 Aug 2026
- **ARTÉA Phase 3 Project Charter draft.** Provided project text, 14 Aug 2026
- Godot Engine licence — Godot is free and open source under the MIT licence.
- OpenAI GPT-5 mini model pricing — current official token pricing used only to validate that bounded pilot interactions are low-cost; the charter retains a conservative AUD allowance.
- Apple Developer Program — native App Store distribution requires a USD $99 annual membership; excluded from the baseline web budget.

FORMULAS Net Benefit = Monetised Benefits − Sponsor Cash Costs. Benefit–Cost Ratio = Monetised Benefits ÷ Sponsor Cash Costs. All dollar figures are indicative and must be confirmed with the sponsor and academic supervisor before charter approval.